

Competency

- To solve the related problem based on trigonometric ratios.

Learning Outcomes

- To introduce trigonometric ratios
- To verify the fundamental trigonometric ratios from the right angled triangle for the given reference angle
- To find the trigonometric ratios of some standard angles
- To solve the right angled triangle by using trigonometric ratios

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define trigonometry - To tell the six trigonometric ratios - To identify the trigonometric ratio based on the reference angle and given sides - To tell the trigonometric ratios of some standard angles
2.	Understanding (U)	- To find the trigonometric ratios of given reference angle in terms of sides of the right angled triangle - To convert a trigonometric ratio in to another - To find the value of expression involving trigonometric ratios with standard angles - To find the unknown length of side of triangle when reference angle is a standard angle
3.	Application (A)	- To verify the fundamental trigonometric ratios from the right angled triangle for the given reference angle - To find the trigonometric ratios of some standard angles from right angled triangle - To solve the right angled triangle by using trigonometric ratios
4.	High Ability (HA)	- To prepare a report on investigating the application of trigonometry, ratios and identities

Required Teaching Materials/ Resources

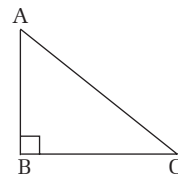
Colourful chart-paper, models of right angled triangles, geometric instruments, colourful markers, chart paper with trigonometric ratios of standard angles up to 900, ICT tool etc

Pre-knowledge: right angled triangles, Pythagorean triplets and name of parts its sides

Teaching Activities

- Recall Pythagorean triplets with examples
- Show a right angled triangle and ask to identify its hypotenuse, base and perpendicular
- Discuss about perpendicular, hypotenuse and base in right angled triangle with respect to one of the acute angle as the reference angle
- Ask the students to make the possible ratios of perpendicular (p), base (b) and hypotenuse (h) then explain about following six trigonometric ratios

(i) $\sin\theta = \frac{p}{h}$	(ii) $\cos\theta = \frac{b}{h}$	(iii) $\tan\theta = \frac{p}{b}$
(iv) $\operatorname{cosec}\theta = \frac{h}{p}$	(v) $\sec\theta = \frac{h}{b}$	(vi) $\cot\theta = \frac{b}{p}$
- Divide the students in to 5 groups. Give them the right angled triangles with different names and ask to identify the perpendicular, base and hypotenuse. Also, ask them to express all six trigonometric ratios in terms of sides of the triangles
- With discussion, give guidelines to solve the problems given in the textbooks.
- Discuss on establishing the relationships of trigonometric ratios
- With discussion, explain the trigonometric ratios of some standard angles up to 90°
- Tell the students to draw the table of trigonometric ratios of standard angles on a chart paper and past on the mathematics corner of the classroom or mathematics lab as project work.
- Encourage the students to evaluate the trigonometric expressions involving the standard angles
- Discuss on the solution of right angles triangles which is supportive for the next class to solve the problems related to height and distance



Solution of selected problems from Vedanta Excel in Mathematics

- 1) If $\sin(90^\circ - \alpha) = \frac{BC}{CA}$, write down the ratio of $\sin\alpha$.

Solution:

Here,

$$\sin(90^\circ - \alpha) = \frac{BC}{CA} = \frac{p}{h}$$

$\therefore$ For reference angle $(90^\circ - \alpha)$

Perpendicular (p) = BC and hypotenuse (h) = CA

Also, $\angle C = 90^\circ - (90^\circ - \alpha) = \alpha$

For reference angle α , perpendicular (p) = AB, base (b) = BC and hypotenuse (h) = CA

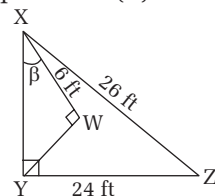
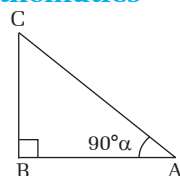
$$\therefore \sin\alpha = \frac{p'}{h'} = \frac{AB}{CA}$$

- 2) From the adjoining figure, show that $\tan\beta = \frac{4}{3}$.

Solution:

- i) In rt. angled $\triangle XYZ$.

$$XY = \sqrt{XZ^2 - YZ^2} = \sqrt{26^2 - 24^2} = 10 \text{ ft}$$

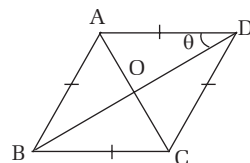


ii) In rt. angled ΔXYZ ,

$$WY = \sqrt{XY^2 - XW^2} = \sqrt{10^2 - 6^2} = 8 \text{ ft}$$

$$\therefore \tan \beta = \frac{WY(p)}{XW(b)} = \frac{8 \text{ ft}}{6 \text{ ft}} = \frac{4}{3}$$

3) In the adjoining figure $ABCD$ is a rhombus in which $AC = 6 \text{ cm}$, $BD = 8 \text{ cm}$ and $\angle ADC = \theta$. Find the values of $\sin \theta$ and $\tan \theta$.



Solution:

i) $OA = \frac{1}{2}AC = \frac{1}{2} \times 6 \text{ cm} = 3 \text{ cm}$

$$OD = \frac{1}{2}BD = \frac{1}{2} \times 8 \text{ cm} = 4 \text{ cm}$$

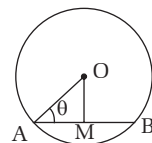
and $\angle AOD = 90^\circ$

ii) In rt.angled ΔAOD ; $AD = \sqrt{OA^2 + OD^2} = \sqrt{3^2 + 4^2} = 5 \text{ cm}$

$$\therefore \sin \theta = \frac{OA(p)}{AD(h)} = \frac{3}{5} \text{ and } \tan \theta = \frac{OA(p)}{OD(b)} = \frac{3}{4}$$

(Diagonal of rhombus do bisect each other at a right angle)

4) In the circle given alongside, O is the centre, M is the mid - point of chord AB . If $AB = 12 \text{ cm}$ and $CM = 8 \text{ cm}$ and $\angle OAM = \theta$, find the values of $\sin \theta$ and $\cos \theta$.



Solution:

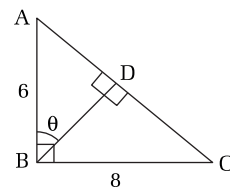
i) $AM = \frac{1}{2}AB = \frac{1}{2} \times 12 \text{ cm} = 6 \text{ cm}$

ii) $OM \perp AB$ [OM joins the centre O and the mid - point M of the chords AB.

iii) In rt. angled ΔAOM ; $AO = \sqrt{OM^2 + AM^2} = \sqrt{8^2 + 6^2} = 10 \text{ cm}$

$$\therefore \sin \theta = \frac{OM(p)}{OA(h)} = \frac{8 \text{ cm}}{10 \text{ cm}} = \frac{4}{5}, \cos \theta = \frac{AM}{OA} = \frac{6 \text{ cm}}{10 \text{ cm}} = \frac{3}{5}$$

5) In the adjoining figure, $AB = 6$, $BC = 8 \text{ cm}$, $\angle ABC = 90^\circ$, $BD \perp AC$ and $\angle ABD = \theta$. Find the value of $\sin \theta$.



Solution:

i) In rt. angled ΔABC , $AC = \sqrt{AB^2 + BC^2} = \sqrt{6^2 + 8^2} = 10$

ii) In ΔABC and ΔABD ,

(a) $\angle ABC = \angle BDA$ [Both are right angles]

(b) $\angle BAC = \angle BAD$ [Common angle]

(c) $\angle ACB = \angle ABD$ [Remaining angles]

$$\therefore \Delta ABC \sim \Delta ABD$$

iii) $\frac{AC}{AB} = \frac{BC}{BD} = \frac{AB}{AD}$ [corresponding angles of similar angle]

$$\text{or, } \frac{10}{6} = \frac{8}{BD} = \frac{6}{AD}$$

From 1st and 2nd ratios, we get

$$\frac{10}{6} = \frac{8}{BD} \quad \therefore BD = \frac{24}{5}$$

From 1st and 3rd ratios, we get

$$\frac{10}{6} = \frac{6}{AD} \quad \therefore AD = \frac{18}{5}$$

iv) In 1st angled $\triangle ABD$,

$$\sin \theta = \frac{AD(p)}{AB(h)} = \frac{18/5}{4} = \frac{3}{5}$$

6) In the given $\triangle ABC$, $\angle B = 90^\circ$, and $\angle CAB = 45^\circ$. If $AB = x$ cm. Prove that $\sin 45^\circ = \frac{1}{\sqrt{2}}$

Solution:

Given: In $\triangle ABC$, $\angle B = 90^\circ$, and $\angle ACB = 45^\circ$, $AB = x$ cm

To prove: $\sin 45^\circ = \frac{1}{\sqrt{2}}$

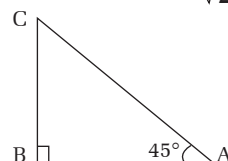
Proof:

i) $\angle ACB = 180^\circ - (\angle A + \angle B) = 180^\circ - (45^\circ + 90^\circ) = 45^\circ$

ii) $AB = BC$ [$\angle A = \angle C = 45^\circ$]

iii) $AC = \sqrt{AB^2 + BC^2} = \sqrt{x^2 + x^2} = \sqrt{2}x$ cm [By using pythagoras theorem]

iv) In rt.angled $\triangle ABC$; $\sin 45^\circ = \frac{BC}{AC} = \frac{x}{\sqrt{2}x} = \frac{1}{\sqrt{2}}$



7) In the adjoining equilateral $\triangle ABC$, $AD \perp BC$ and $AC = 2a$ units.

Prove that $\sin 60^\circ = \frac{\sqrt{3}}{2}$

Solution:

Given: $\triangle ABC$ is an equilateral triangle. $AD \perp BC$ and $AC = 2a$ units.

To prove: $\sin 60^\circ = \frac{\sqrt{3}}{2}$

Proof:

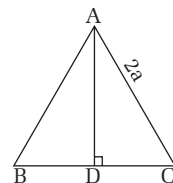
i) $\angle C = 60^\circ$ [Being an angle of equilateral triangle]

ii) $CD = \frac{1}{2} BC = \frac{1}{3} \times 2a = a$ [Median of equilateral $\triangle$ bisects its base]

iii) In rt. $\triangle ADC$; $AD = \sqrt{AC^2 - CD^2} = \sqrt{(2a)^2 - a^2} = \sqrt{3}a$ cm [By Pythagorom theorem]

iv) $\sin 60^\circ = \frac{AD(p)}{AC(h)} = \frac{\sqrt{3}a}{2} = \frac{\sqrt{3}}{2}$

$\therefore \sin 60^\circ = \frac{\sqrt{3}}{2}$ proved

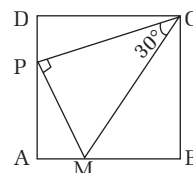


8) In the given figure, $ABCD$ is a square and $\angle CPM = 90^\circ$, if $CM = 4$ cm, $PD = \sqrt{3}$ cm. Find the side of square.

Solution:

i) In rt. angled $\triangle CMP$; $\cos 30^\circ = \frac{PC(b)}{CM(h)} = \frac{\sqrt{3}}{2} = \frac{CP}{4}$
 $\therefore CP = 2\sqrt{3}$ cm

ii) In rt. angle $\triangle PCD$; $CD = \sqrt{CP^2 - DP^2} = \sqrt{(2\sqrt{3})^2 + (\sqrt{3})^2} = 3$ cm
Hence, the side of square $ABCD$ is 3 cm.



- 9) In the given figure, ABCD is a rectangle and ADE is a right angle triangle. Find the size of EC.

Solution:

i) In rectangle ABCD, $AB = CD = 50\sqrt{3}$ m

ii) In rt. $\triangle ABC$; $\tan 45^\circ = \frac{p}{b} = \frac{AB}{BC}$

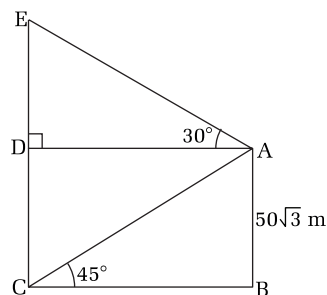
or, $1 = \frac{50\sqrt{3} \text{ m}}{BC} \therefore BC = 50\sqrt{3} \text{ m}$

iii) $AD = BC = 50\sqrt{3} \text{ m}$
[Opposite sides of rectangle ABCD]

iv) In rt. $\triangle EDA$. $\tan 30^\circ = \frac{p}{b} = \frac{ED}{AD}$

or, $\frac{1}{\sqrt{3}} = \frac{ED}{50\sqrt{3} \text{ m}} \therefore ED = 50 \text{ m}$

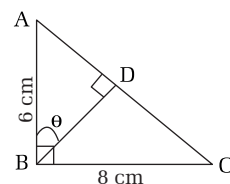
Hence, $EC = ED + DC = 50 \text{ m} + 50\sqrt{3} \text{ m} = 136.6 \text{ m}$



Extra Question

1. In the adjoining figure, $AB = 6$ cm, $BC = 8$ cm, $\angle ABC = 90^\circ$, $BD \perp AC$ and $\angle ABD = \theta$, find the value of $\cos\theta$ and $\tan\theta$.

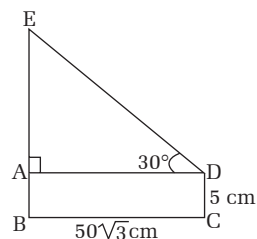
Ans: $\frac{4}{5}, \frac{3}{4}$



2. Prove that: $\frac{1 - \tan 30^\circ}{1 + \tan 30^\circ} = \frac{1 - \sin 30^\circ}{1 + \sin 30^\circ}$

3. In the given figure, ABCD is a rectangle and EAD is a right angled triangle. Find the length of BE.

Ans: 55 cm



4. In the equilateral triangle ABC, $AD \perp BC$ and $AC = x$ cm. Show that:

$\sin 60^\circ = \frac{\sqrt{3}}{2}$ and $\tan 30^\circ = \frac{1}{\sqrt{3}}$.

